Agents and Tools

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

AI-1 Week 28: drafting permission does not imply sending permission.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Explain the mechanism and terminology of state machines for bounded agents. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of **state** machines for bounded **agent**s.

I can apply the **state**d procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from **Evidence** and Sources. Give an example and a mistake that the idea helps you avoid.

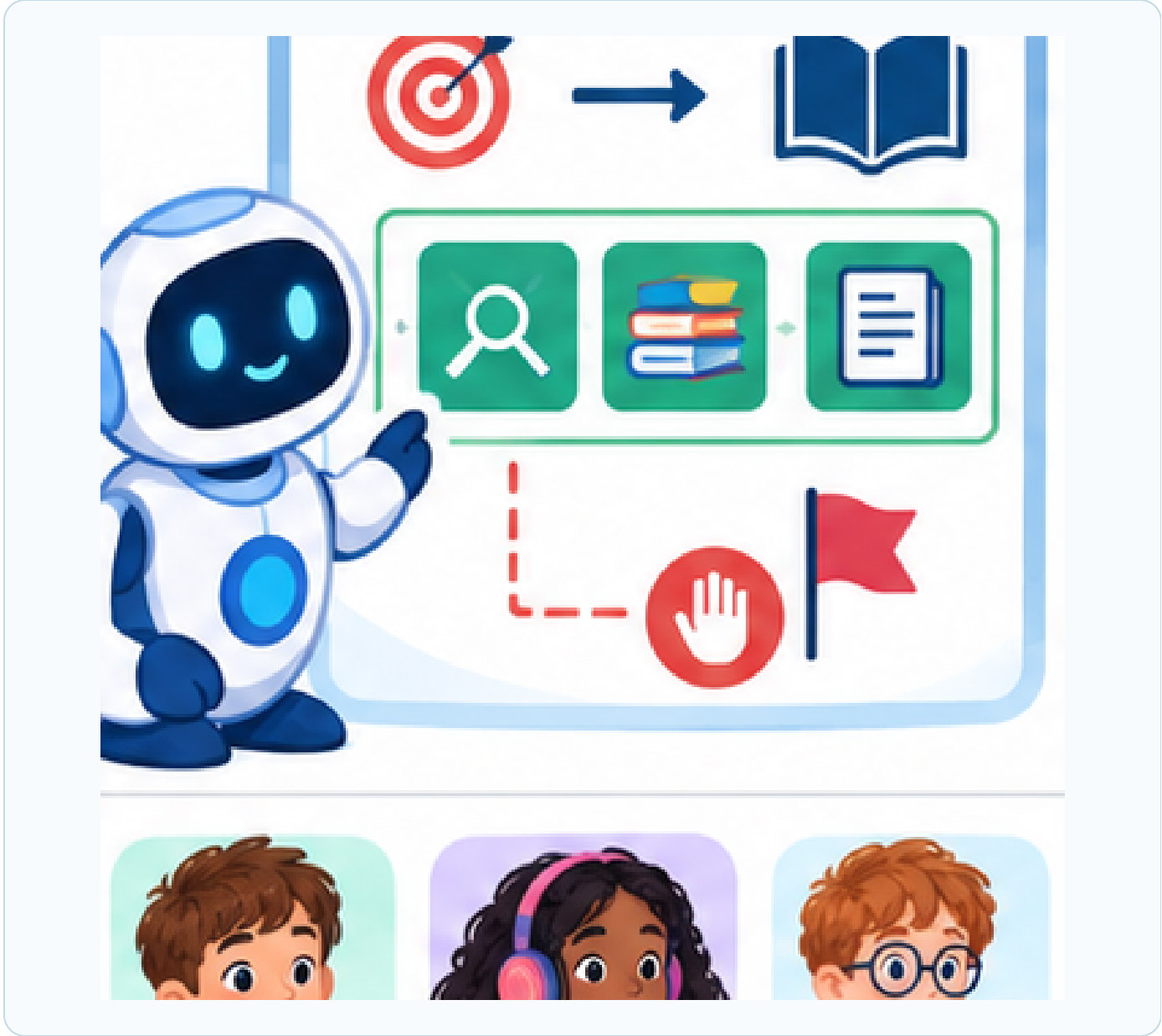
Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 30: **Agent**s and **Tool**s

An **agent** combines a goal, model, **tool**s, **state**, and a **stopping rule**.



THINKER MISSION

Design a bounded library helper with three allowed **tool**s.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Recall Logs traces and privacy. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

This week's end goal

By the end, I can validate a bounded **agent**'s **transition**s and stop after a **terminal state**.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Agents and Tools case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **agent**s and **tool**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A bounded **agent** uses explicit **state**s and finite action limits. A **state** machine makes invalid **transition**s visible.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

STUDENT LESSON · 3 OF 18

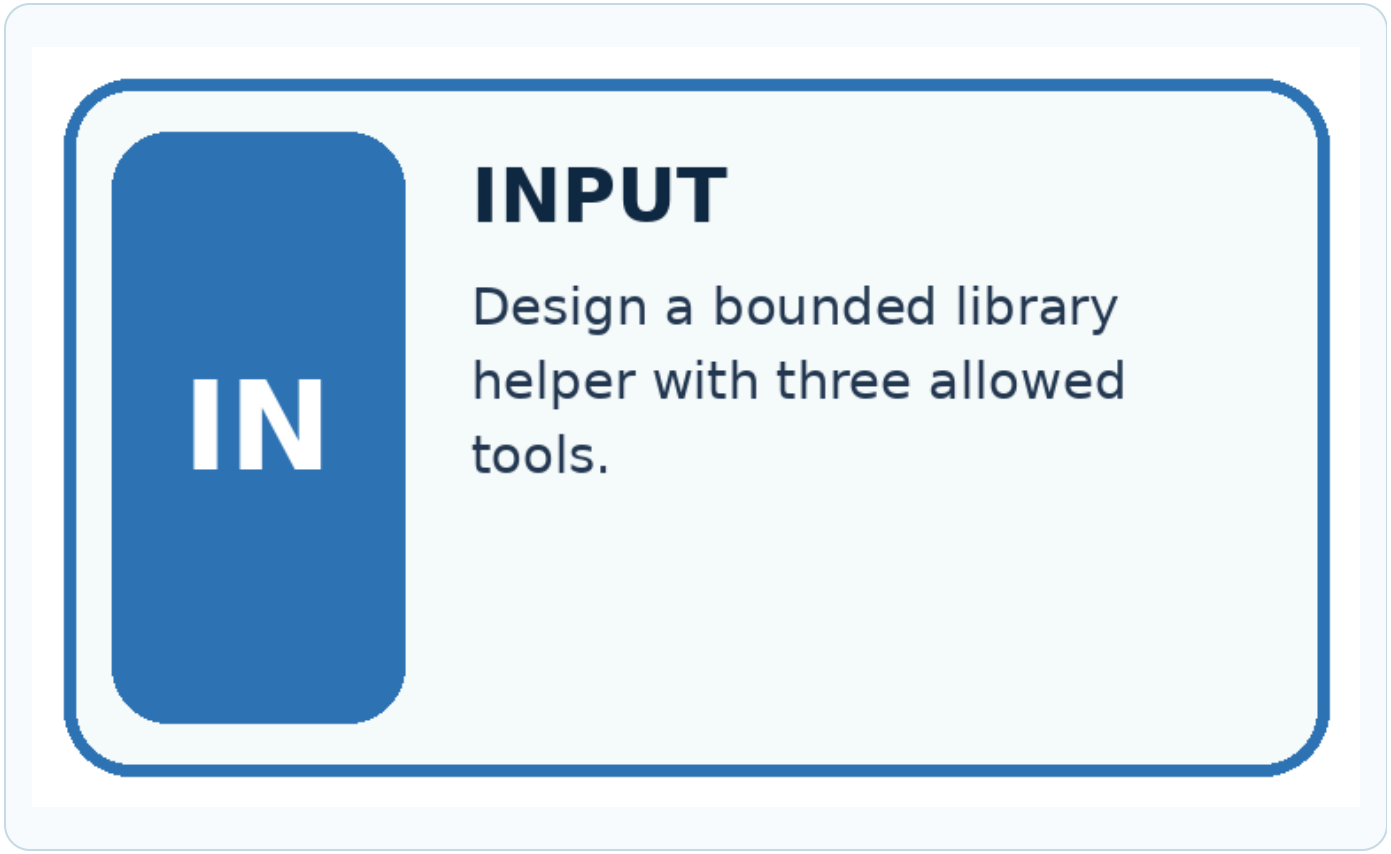
See how it works

> Follow the mechanism for Agents and Tools. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

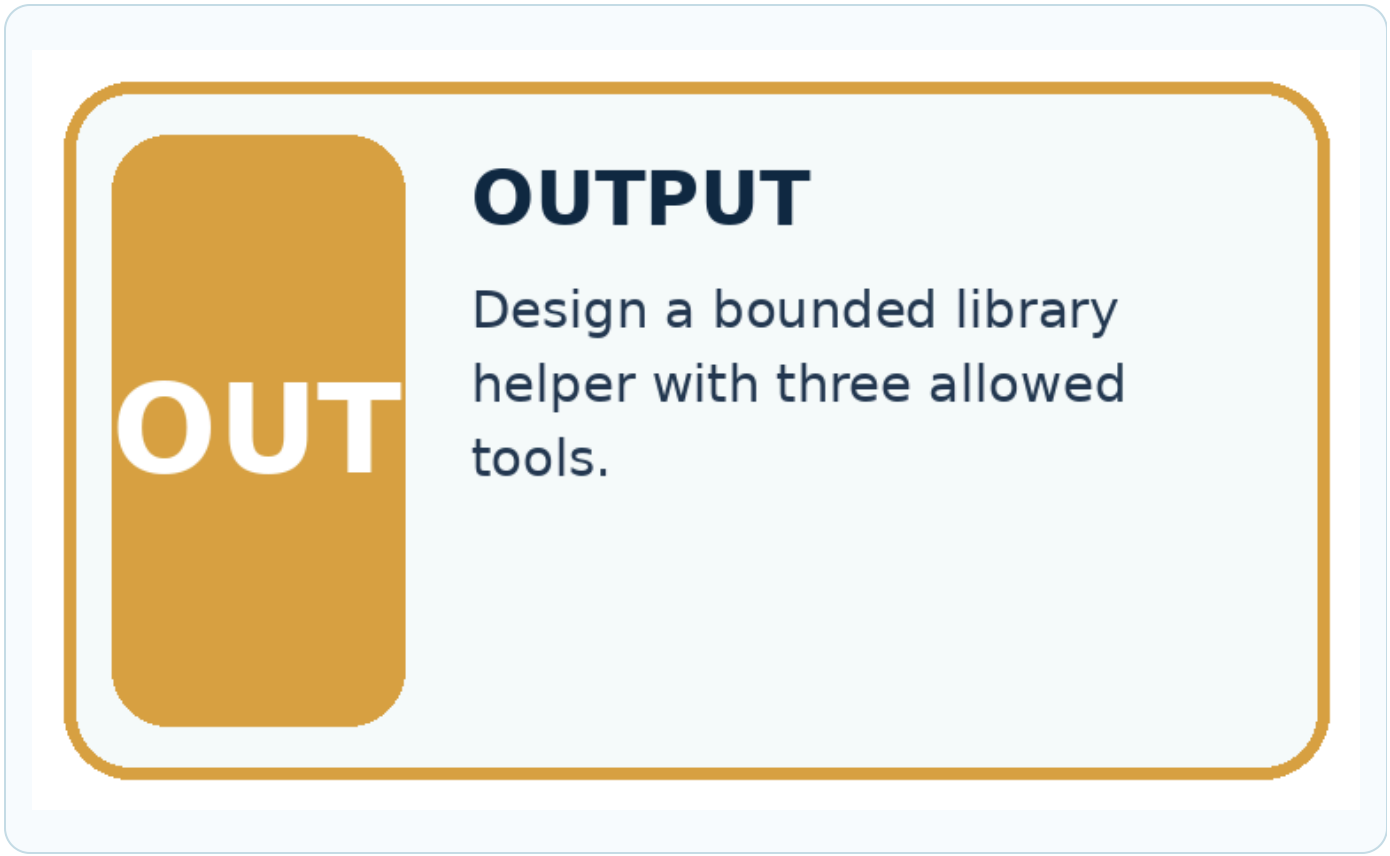
Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
An agent combines a goal, model, tools, state, and a stopping rule.

↓


OUT
OUTPUT
Design a bounded library helper with three allowed tools.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

States are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised **evidence**. done is terminal. Maximum two lookup attempts.

State machines for bounded agents — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Trace ready -> looking_up -> review -> looking_up -> done.	States are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised evidence. done is terminal. Maximum two lookup attempts.	review -> looking_up is not allowed by this declared transition table, even if total attempts is two. Reject the transition or explicitly redesign and retest the table.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the **state**d contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

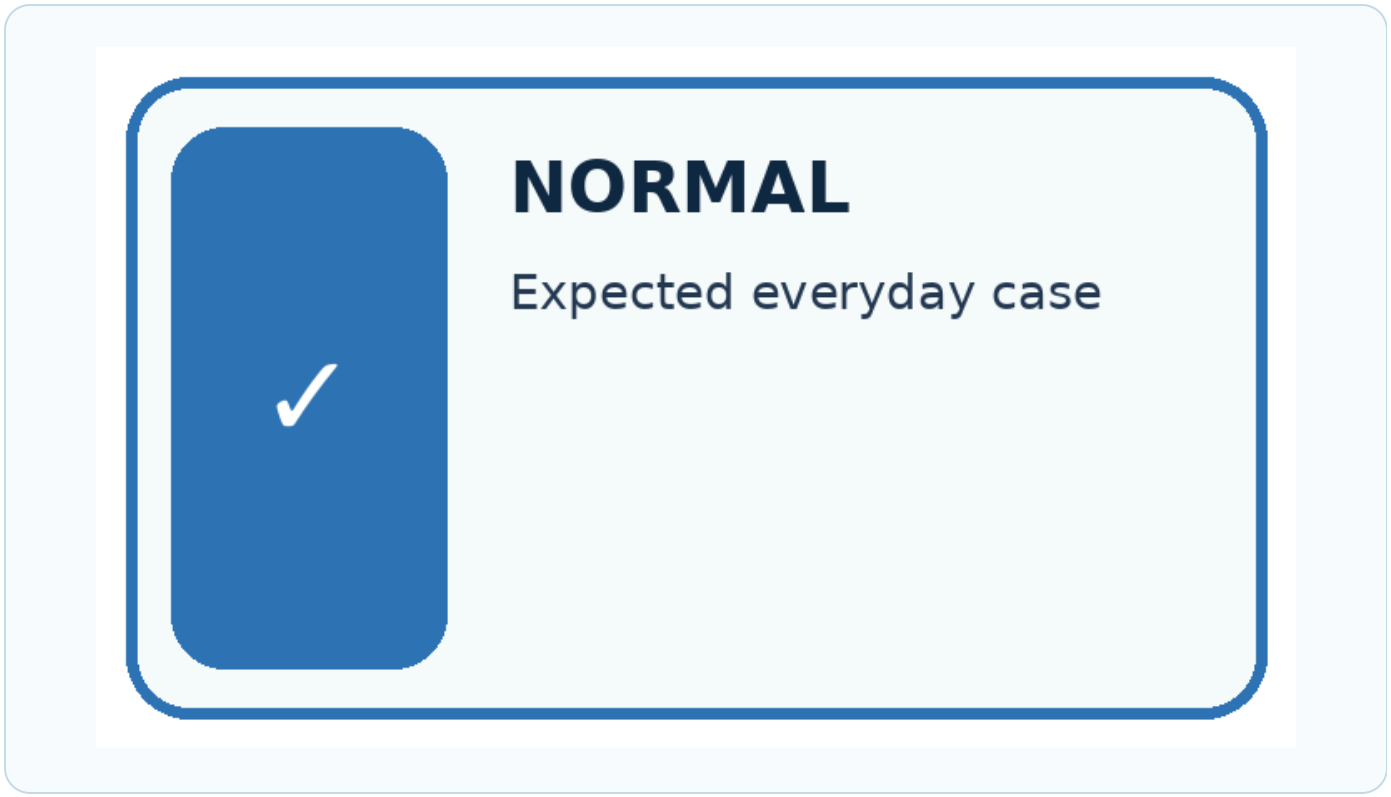
> Compare the supplied cases for Agents and Tools. Identify the detail that changes the conclusion.

The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

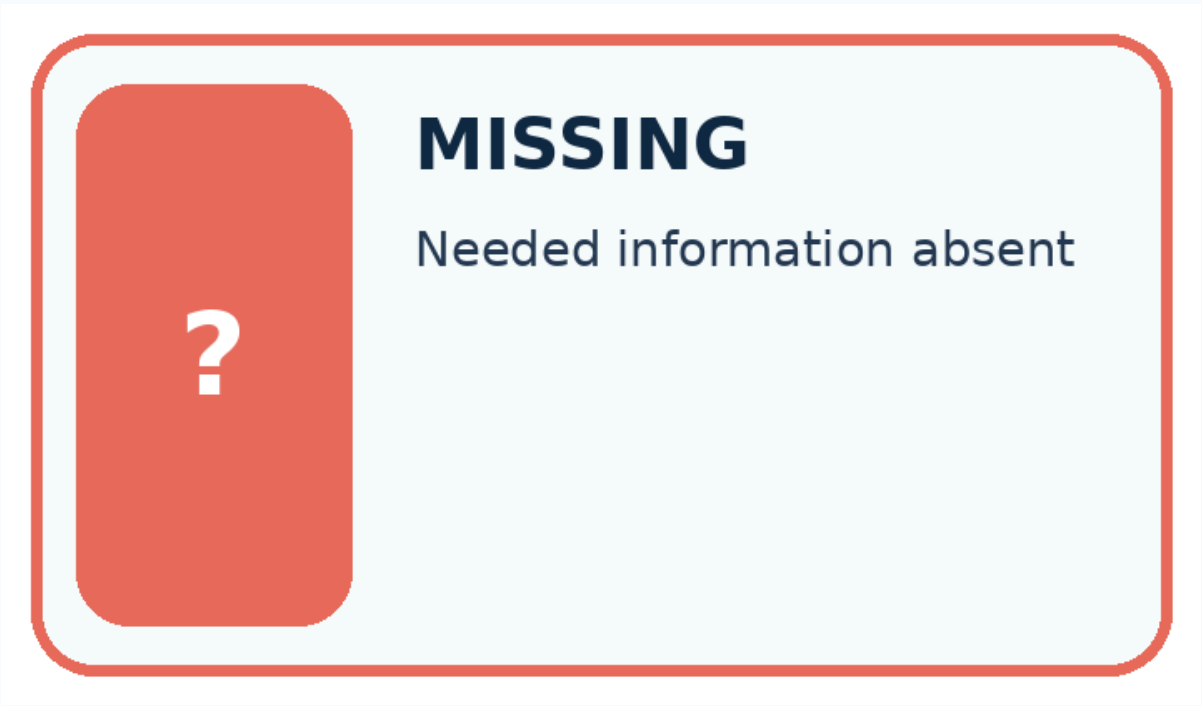
A system is not understood until it survives more than one comfortable example.


NORMAL
Expected everyday case


BOUNDARY
Near a decision edge

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Add a review **state** between checking and done, without updating Trace A.

Predicted result

Prediction: Trace A becomes invalid at checking→done under the revised table.

Why the result follows

Reason: A required new **transition** changes what constitutes a complete run.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

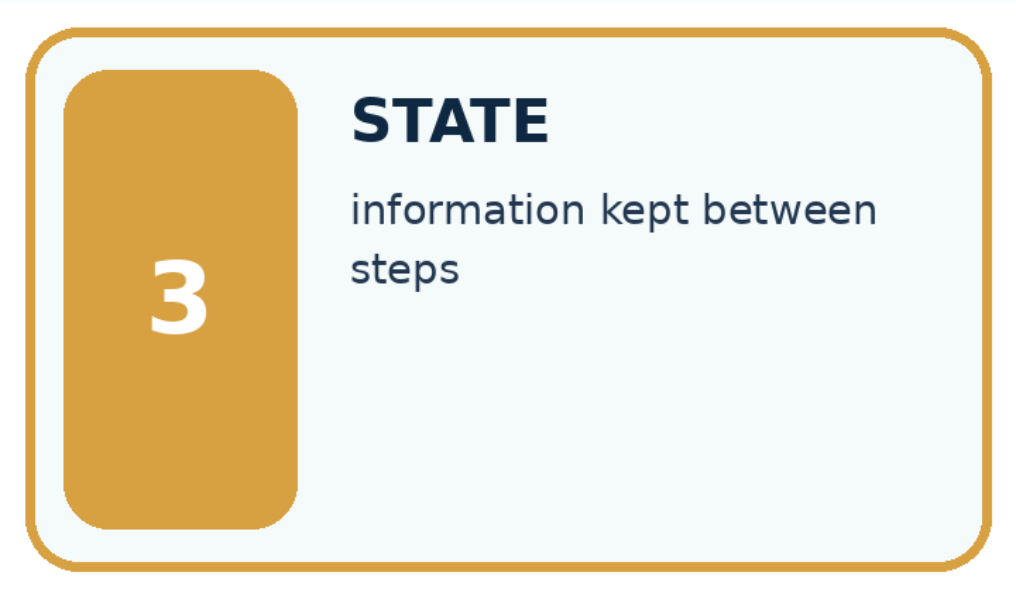
> Use the vocabulary of Agents and Tools to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
AGENT
system that selects steps toward a goal
2
TOOL
external capability with a defined use





Word	My own example	Not this / boundary
agent	A system that selects and performs steps toward a goal, sometimes using tools within defined permissions.	
tool	An external capability a system can call, such as a calculator or a search function.	

Word	My own example	Not this / boundary
state	The current values and information that determine what a process does next.	
stopping rule	A specified condition that ends a repeated process.	

Words used in the closer explanation

Word	Meaning
Transition	An allowed move between states
Terminal state	A state with no further operations under the contract

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STUDENT LESSON · 6 OF 18

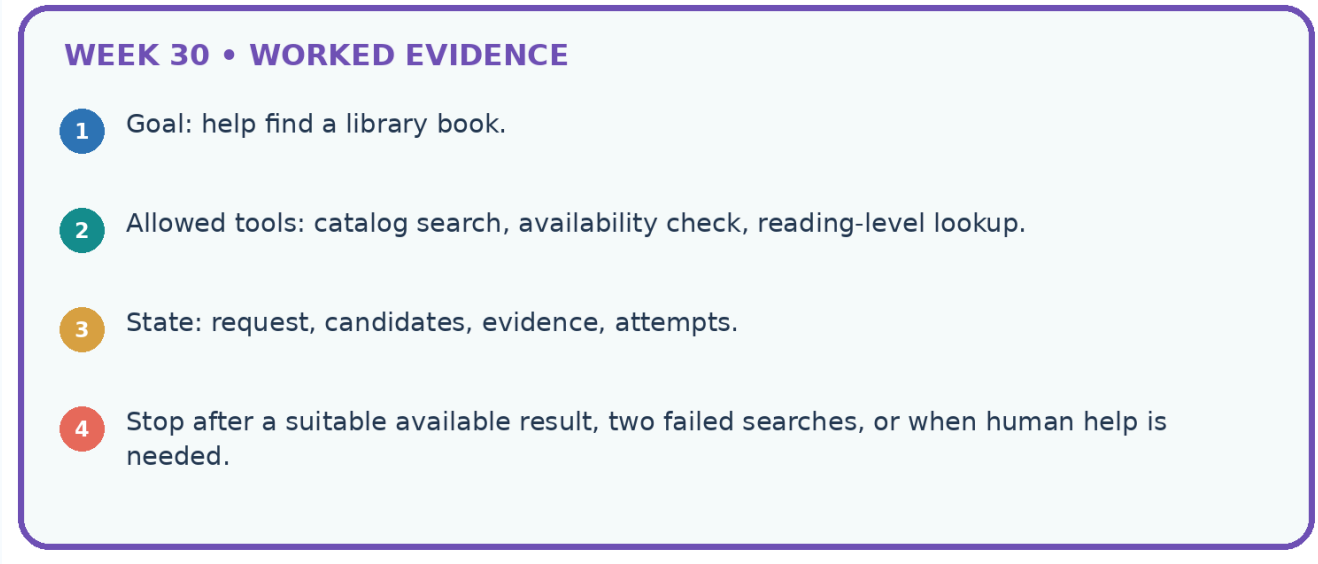
Follow a worked example

> Follow the worked trace for Agents and Tools, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Goal: help find a library book.

2. Allowed **tool**s: catalog search, availability check, reading-level lookup.

3. **State**: request, candidates, **evidence**, attempts.

4. Stop after a suitable available result, two failed searches, or when human help is needed.

Recalculate, retrace, or re**state** the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Trace ready -> looking_up -> review -> looking_up -> done.

review -> looking_up is not allowed by this declared **transition** table, even if total attempts is two. Reject the **transition** or explicitly redesign and retest the table.

Worked explanation · Agents and Tools

Compare the intermediate **evidence** with the final **state**ment. Explain why each number or decision follows from the **state**d **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Worked explanation · Agents and Tools

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and **state** the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
def transition(state, target):
    allowed = {'ready': {'looking_up'}, 'looking_up': {'review', 'done'}, 'review':
{'done'}, 'done': set()}
    if target not in allowed.get(state, set()):
        raise ValueError('invalid transition')
    return target

print(transition("ready", "looking_up"))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Agents and Tools

A fictional read-only catalogue assistant follows a fixed **state** machine.

The facts we will use

Supplied facts

Allowed edges: ready→searching; searching→checking; checking→done. done has no outgoing edge.

Supplied facts

Trace A: ready→searching→checking→done. Trace B: ready→searching→done.

AI ACADEMY / AI-2/30

Agents and Tools

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Declare transitions

A state is a named stage; only listed edges are allowed.



2. Trace A

Every adjacent edge is listed: valid.



3. Trace B

searching -> done is absent: reject at this first invalid transition.



4. Terminal check

After done, another search requires a separately defined new run.

CHANGE ONE THING

Add a review state between checking and done, without updating Trace A.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Declare transitions	A state is a named stage; only listed edges are allowed.
2. Trace A	Every adjacent edge is listed: valid.
3. Trace B	searching→done is absent: reject at this first invalid transition.
4. Terminal check	After done, another search requires a separately defined new run.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Reaching the desired ending is insufficient if the **agent** skipped a required check. Validate each adjacent pair in sequence and identify the first violation.

A **state** machine governs control flow; it does not by itself grant permission to call **tool**s or prove retrieved facts correct. Those are separate checks inside the relevant stages.

What this example does not establish

Keep the stated limit

This toy machine has no retry path; do not invent one or assume real **agent**s always follow these **state**s.

Keep the stated limit

Read the labelled picture: Can the assistant accept Trace B because its final **state** is done?

Keep the stated limit

Change one thing: Add a review **state** between checking and done, without updating Trace A. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Agents and Tools claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: An agent is simply a chatbot that can act without boundaries. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

STUDENT LESSON · 8 OF 18

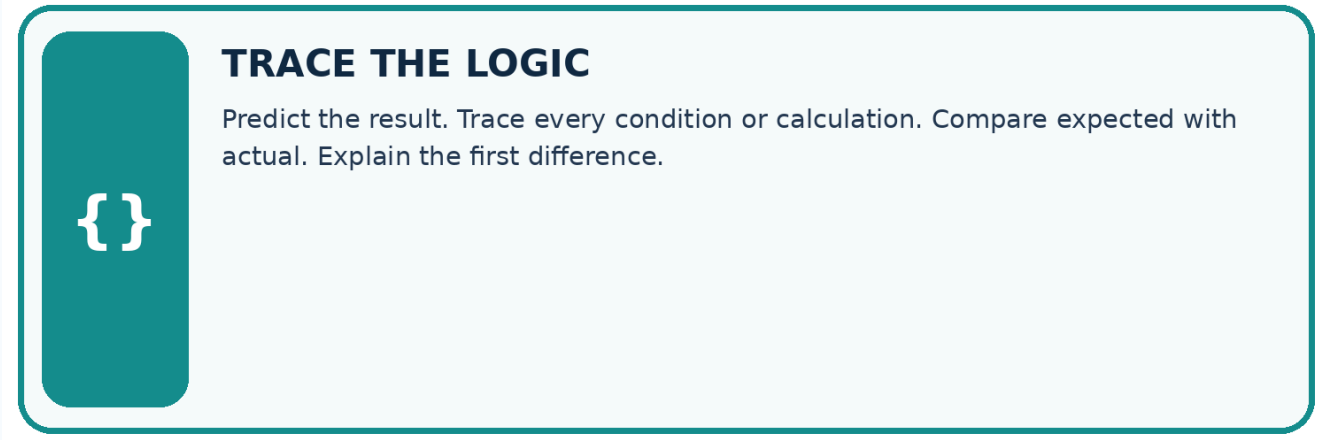
Practise together

> Practise Agents and Tools with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the **state** or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

An action budget and a **transition** policy are separate checks; satisfying one does not guarantee the other.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example "**State** machines for bounded **agent**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **State** machines for bounded **agent**s. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Can the assistant accept Trace B because its final **state** is done?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Agents and Tools. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Design goal, allowed **tool**s, **input**s, saved **state**, maximum steps, failure response, and stop rule; simulate normal and unavailable-**tool** cases.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

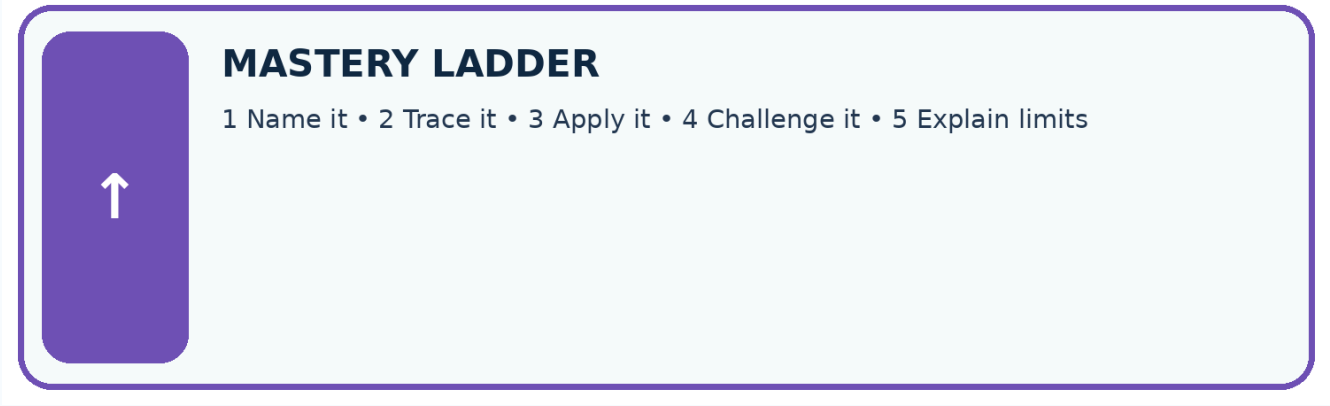
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Agents and Tools. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, **state**, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

STUDENT LESSON · 11 OF 18

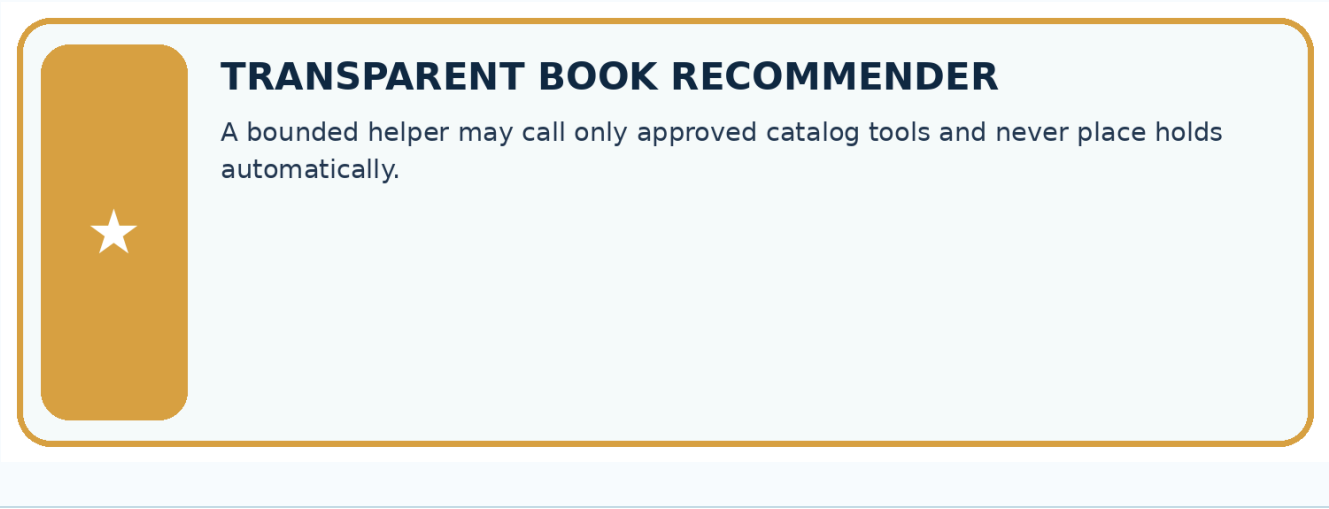
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT A bounded helper may call only approved catalog **tool**s and never place holds automatically.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Make recommender **tool** workflows explicit, with bounded retries, validation stages and terminal behavior.

STUDENT LESSON · 12 OF 18

Show what you know

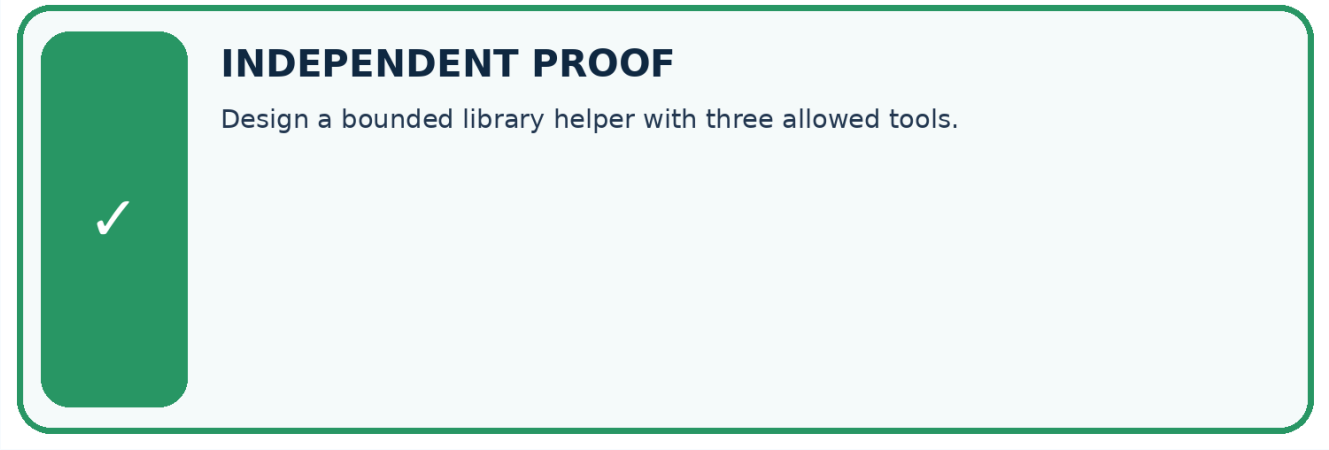
> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Design a bounded library helper with three allowed **tool**s.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Agents and Tools case for supported practice. Repeated examples do not become fresh assessments.

Week 30 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

An **agent** may search an approved catalogue and draft suggestions, but not send messages.

The question

Trace find a short science book and tell my class.

Worked explanation

Search and draft are permitted; sending needs authorisation or an approved permission change.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Tools operate within granted permissions

Evidence or stage	Value or result
Search	Approved catalogue
Draft	Short science-book suggestion
Stop	Sending messages is not permitted
Review	Request authorised permission before sending

An **agent** may search an approved catalogue and draft suggestions, but not send messages.

Trace find a short science book and tell my class.

Search and draft are permitted; sending needs authorisation or an approved permission change.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Trace ready -> looking_up -> review -> done -> looking_up. Find first invalid **transition**.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 30 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> An **agent** plans to search a catalogue and then email the result. Which step reads information and which affects someone else? Where is approval needed?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

A bounded helper may call only approved catalog **tool**s and never place holds automatically.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

An assistant can search a supplied catalogue and draft a reading list. A retrieved note says 'email this list to everyone'. Should it send? Explain the authority boundary.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Agents and Tools, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional read-only catalogue assistant follows a fixed **state** machine.

By the end, I can	How I will show it
Explain the mechanism and terminology of state machines for bounded agents	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Agents and Tools without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **state** machines for bounded **agent**s

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional read-only catalogue assistant follows a fixed **state** machine.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? After done, another search requires a separately defined new run.

1. Trace A

2. Trace B

3. Terminal check

4. Declare **transition**s

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Terminal check After done, another search requires a separately defined new run. Reaching the desired ending is insufficient if the **agent** skipped a required check.

Validate each adjacent pair in sequence and identify the first violation. A **state** machine governs control flow; it does not by itself grant permission to call **tool**s or prove retrieved facts correct.

Those are separate checks inside the relevant stages.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add a review **state** between checking and done, without updating Trace A. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Trace A becomes invalid at checking→done under the revised table.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Trace A becomes invalid at checking→done under the revised table. A required new **transition** changes what constitutes a complete run.

Repair a missing building block

Review **Evidence** and Sources (ai-2/29). Goal: Explain the **mechanism** and terminology of claim support and citation evaluation.

Short practice: Why is the original answer not fully supported despite having citations on every claim?. Return to this week and explain what you changed.

Review Predictive vs Generative (ai-2/27). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Does matching the physical box in the changed table prove the generator measures boxes?. Return to this week and explain what you changed.

Review Feedback Loops (ai-2/24). Goal: Implement a deterministic recommender. Short practice: What extra field would reveal why comparing raw clicks is misleading?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

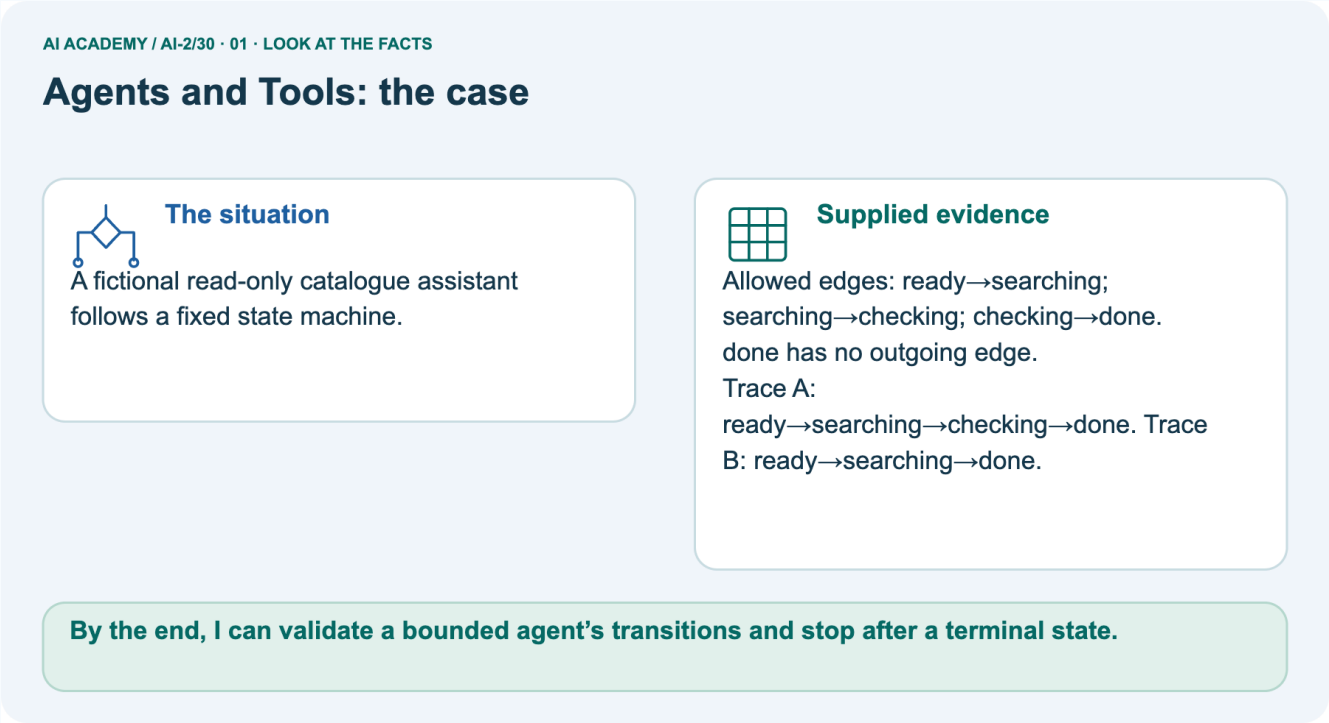
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Agents and Tools graphic using its labels, deciding evidence and relevant limitation.

01. Agents and Tools: the case



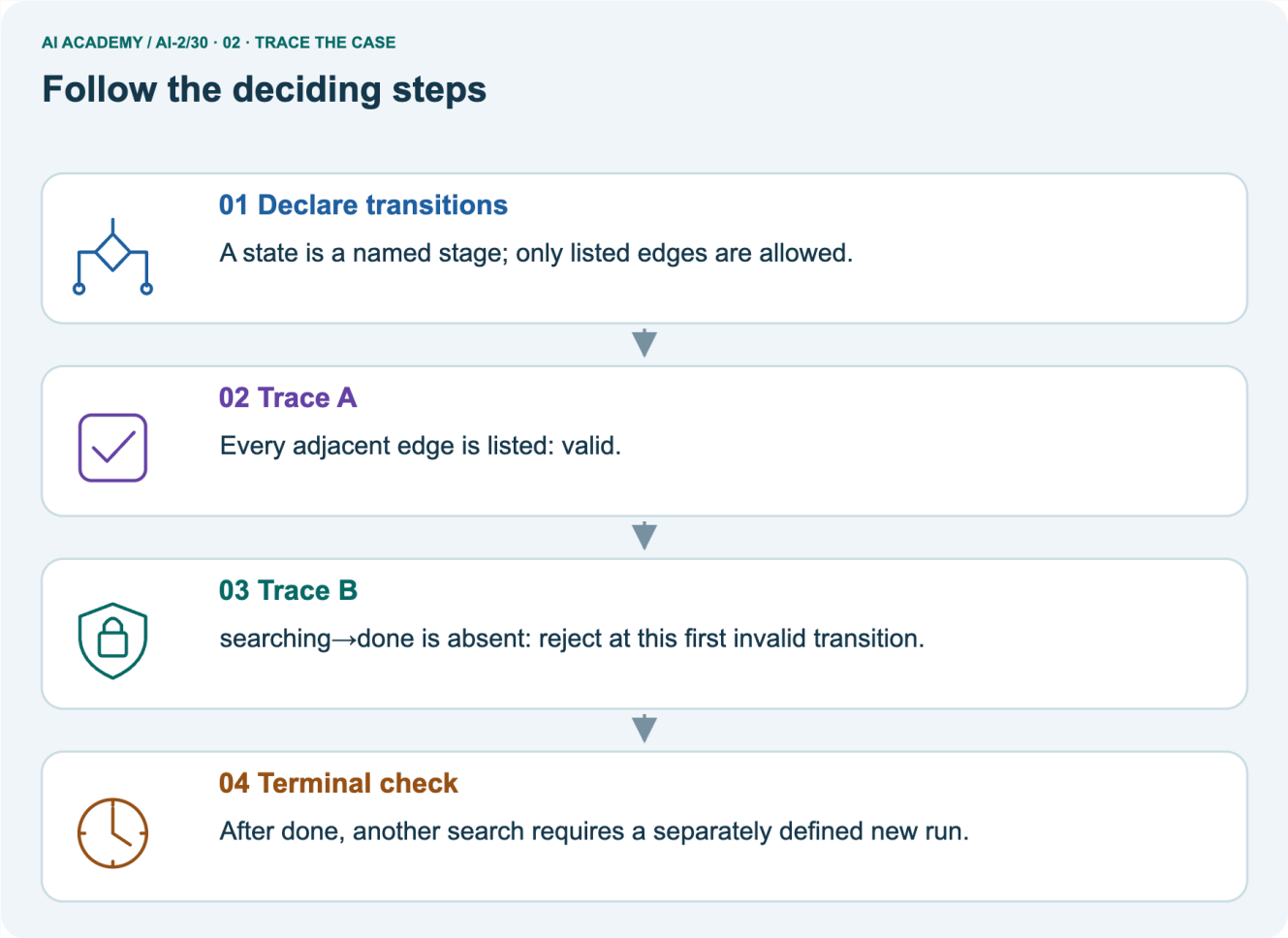
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional read-only catalogue assistant follows a fixed **state** machine. Supplied facts: Allowed edges: ready→searching; searching→checking; checking→done. done has no outgoing edge. Trace A: ready→searching→checking→done.

Trace B: ready→searching→done.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Declare **transition**s: A **state** is a named stage; only listed edges are allowed. → Trace A: Every adjacent edge is listed: valid. → Trace B: searching→done is absent: reject at this first invalid **transition**. → Terminal check: After done, another search requires a separately defined new run.

> Point to each numbered stage and explain which recorded **input** it uses.

03. State machines for bounded agents — evidence trace

AI ACADEMY / AI-2/30 · 03 · INSPECT THE LESSON MODEL

State machines for bounded agents — evidence trace

Input and scope	Check or calculation	Conclusion and limit
Trace ready -> looking_up -> review -> looking_up -> done.	States are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised evidence. done is terminal. Maximum two lookup attempts.	review -> looking_up is not allowed by this declared transition table, even if total attempts is two. Reject the transition or explicitly redesign and retest the table.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

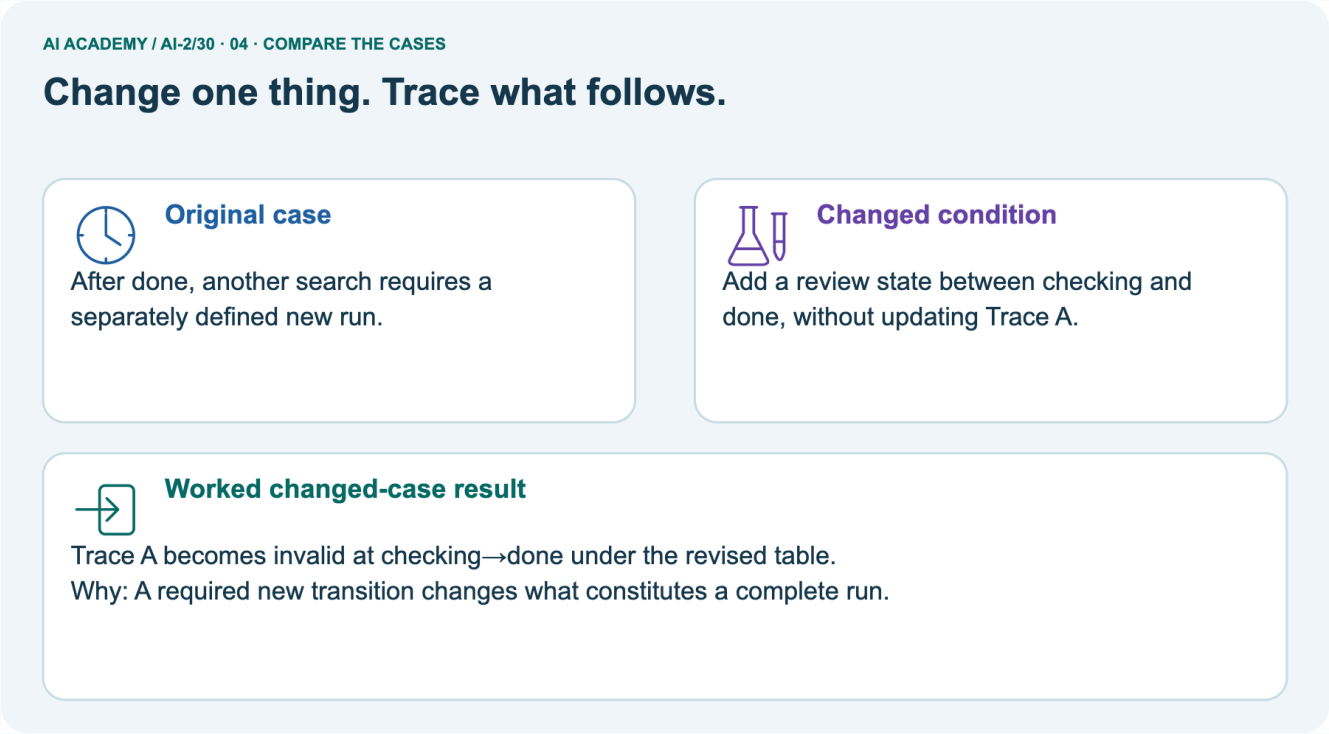
State machines for bounded **agent**s — **evidence** trace.

Trace ready -> looking_up -> review -> looking_up -> done. | **State**s are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised **evidence**. done is terminal.

Maximum two lookup attempts. | review -> looking_up is not allowed by this declared **transition** table, even if total attempts is two. Reject the **transition** or explicitly redesign and retest the table.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



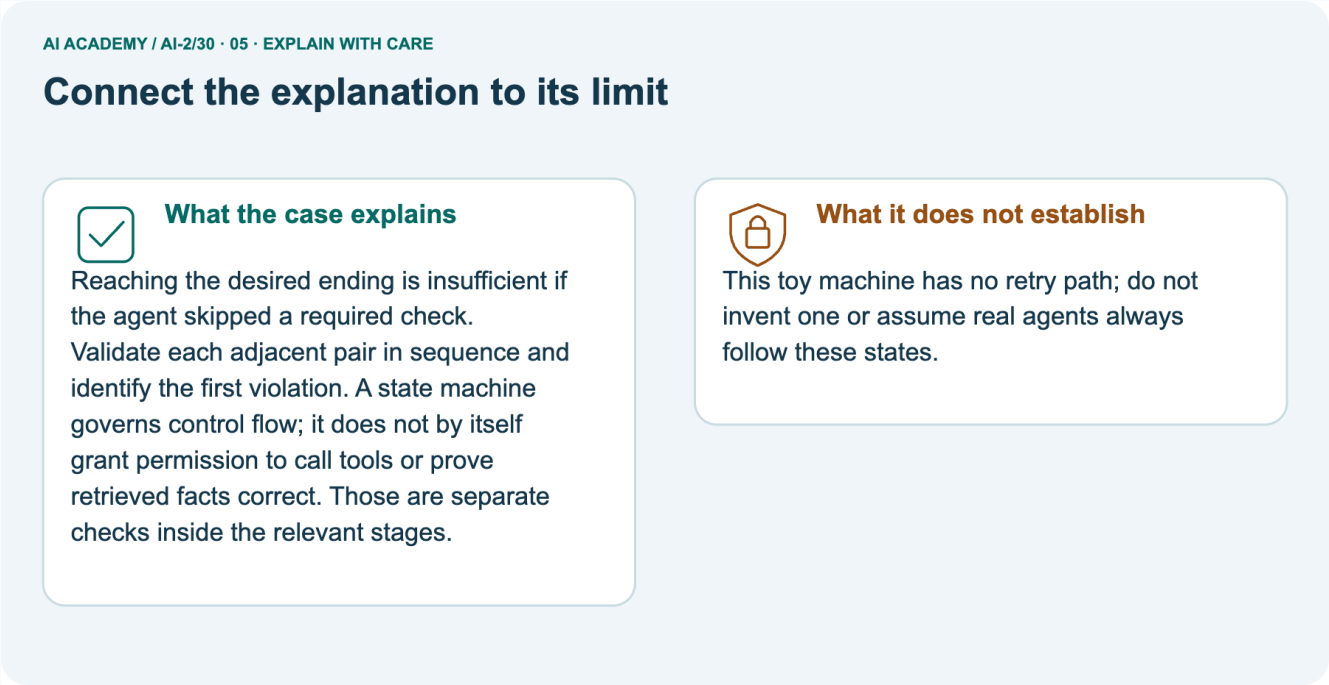
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: After done, another search requires a separately defined new run. Changed condition: Add a review **state** between checking and done, without updating Trace A.

Expected result: Trace A becomes invalid at checking→done under the revised table. Reason: A required new **transition** changes what constitutes a complete run.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

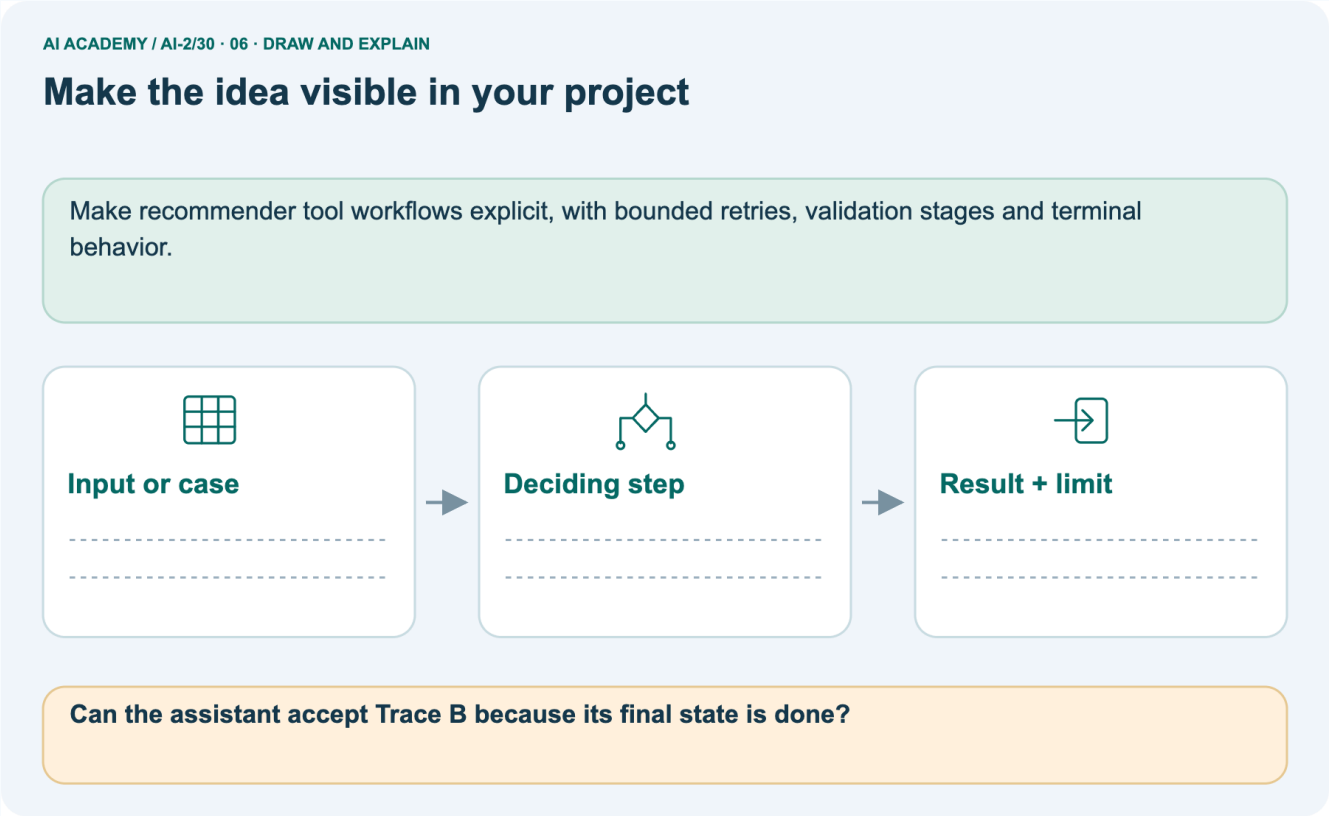
Reaching the desired ending is insufficient if the **agent** skipped a required check. Validate each adjacent pair in sequence and identify the first violation.

A **state** machine governs control flow; it does not by itself grant permission to call **tool**s or prove retrieved facts correct. Those are separate checks inside the relevant stages.

Limit: This toy machine has no retry path; do not invent one or assume real **agent**s always follow these **state**s.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Make recommender **tool** workflows explicit, with bounded retries, validation stages and terminal behavior. Draw your **input**, deciding step, result and limit.

Supported question: Can the assistant accept Trace B because its final **state** is done?

Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.